Project 5: Full Box–Jenkins ARMA study with forecasting

Identification · Estimation · Diagnostics · Forecast

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Context

This project consolidates all of Chapters 3–5: stationarisation, ACF/PACF identification, ARMA estimation, residual diagnostics and forecasting. You will produce a complete Box–Jenkins study on the Souvenir Sales data, on the differenced log-scale, where an ARMA model is appropriate.

The deliverable serves two purposes: it will be re-used in Chapter 6 (ARCH/GARCH) where you will check the residuals for volatility clustering.

Deliverables

- Reproducible `project5_<name>.py` with seed 42.
- Report PDF (8–10 pages) including all figures, all numerical outputs, and a final back-transformed forecast with 95% CI.
- Optional 10-min oral defence.

Exercise 1 – Stationarisation

(1) Load `data/SouvenirSales.csv`. Compute $y_t = \log(\text{sales}_t)$.
(2) Apply the seasonal log-difference $w_t = (1 - L)(1 - L^{12})y_t$. Verify by ADF/KPSS that w_t is stationary.
(3) From now on, model w_t as an $ARMA$ on the stationary scale. Plot w_t, its sample ACF and PACF up to lag 36.

Exercise 2 – Identification

(1) Read the ACF/PACF of w_t at the regular lags (1–11): which (small) AR or MA orders look plausible?
(2) Read at the seasonal lags $(12, 24, 36)$: does the spike at lag 12 persist or disappear after one differencing?
(3) Propose three candidate orders, e.g. $(0, 1)$, $(1, 0)$, $(1, 1)$ on the regular part (and you may add a seasonal component if needed; see Chapter 4).

Exercise 3 – Estimation and selection

(1) Fit each candidate by MLE (`statsmodels.tsa.arima.ARIMA`). For each, report:

- the estimated coefficients with standard errors;
- the log-likelihood, AIC, BIC, $\hat{\sigma}^2$;
- the Ljung–Box p-value at $m = 12$ and $m = 24$.

(2) Choose the best model on the (AIC, BIC, Ljung–Box) triplet. Justify in 5 lines.
(3) Plot the model residuals: time plot, sample ACF, QQ-plot vs $\mathcal{N}(0,1)$. Are they convincingly white noise?

Exercise 4 – Forecasting on the stationary scale

(1) On the chosen ARMA, produce a 12-step forecast of $w_{T+1}, \ldots, w_{T+12}$ with 95% CI.
(2) Check that the forecast returns to the unconditional mean $\mathbb{E}[w] \approx 0$ as h grows.
(3) Plot the in-sample fit and the forecast with the CI band.

Exercise 5 – Back to the original scale

(1) Recall $w_t = (1-L)(1-L^{12})y_t$. Inverting one step at a time, derive the recursion that takes $\widehat{w}_{T+h}$ back to $\widehat{y}_{T+h}$ (telescope sum on the seasonal and the regular difference).
(2) Implement the back-recursion in Python. Plot y_t together with $\widehat{y}_{T+h}$ and the back-transformed CI.
(3) Take the exponential to recover sales:

$$\widehat{\text{Sales}}_{T+h} = \exp(\widehat{y}_{T+h} + \widehat{\sigma}_h^2/2)$$

(conditional mean) or $\exp(\widehat{y}_{T+h})$ (median). Report the predicted sales for the 12 months following December 2001.

Exercise 6 – Bonus: hold-out validation

(1) Split: train = Jan 1995 – Dec 2000 (72 obs), test = Jan – Dec 2001 (12 obs).
(2) Re-fit the chosen ARMA on the training set (with the same differencing recipe), forecast the test horizon, back-transform.
(3) Report MAE, RMSE, MAPE on the original scale. Discuss: does the model under- or over-estimate the December peak?

Marking grid (out of 100)

Criterion	**Points**
Stationarisation + ADF/KPSS verification	10
ACF/PACF identification of three candidates	15
MLE estimation + AIC/BIC selection	20
Residual diagnostics (time, ACF, QQ, LB)	15
Forecast on the stationary scale + CI	10
Back-transform to the original scale	10
Bonus: hold-out validation (RMSE/MAPE)	10
Reproducibility + report quality	10
Total	**100**

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